GoreeCloud Search Fusion Engine - Development integration contract

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Stage: Development-only experimental source on the GoreeCloud Search development branch. **Not** connected to the deployed website, live external providers, or a public API. An offline, opt-in versioned fusion path is now included in the *candidate* native Linux .pyz; no new service was deployed.

native_search/fusion.py contains the first independent GoreeCloud-owned result-fusion primitive. It accepts immutable SourceResult cards for web, images, videos and news, and emits FusedResult objects. Its standard-library implementation performs no I/O or background requests.

native_search/dispatch.py adds the **Development-only adapter-dispatch contract**. It accepts a bounded query and a finite set of named asynchronous provider adapters, checks their explicit per-category ProviderRights, runs only admitted adapters with cooperative deadlines and a maximum concurrency limit, and combines their normalized cards through the existing ranker. It returns a versioned FusionResponse with one fused result list and query-free per-provider health states (blocked, ok, empty, timeout, failed). It neither ships provider adapters nor authorizes provider use.

native_search/api.py exposes POST /v1/fusion on the Linux-native service's existing 127.0.0.1 listener. The default **Web** category now automatically fuses results from GoreeCloud's own local SQLite FTS5 index through a sealed, built-in goreecloud_index adapter. Its indexed documents remain local and no external provider is configured or contacted by default. Other categories (images, videos, news) still return HTTP 503 from this route when they have no configured providers. A developer must explicitly start the service with --fusion-fixture /absolute/path/to/synthetic.json to admit strictly bounded, local synthetic cards for an offline preview. The response combines source cards in one category and reports only bounded, query-free result cards and provider states; it is not live metasearch.

v1 offline dispatcher boundaries

- Rights default to **deny**. Fetch, fusion and display must each be explicitly allowed and accompanied by a nonempty reference to reviewed upstream terms; category membership is also explicit. rights_basis defaults to unverified and cannot admit. synthetic_fixture admits only with the explicit allow_synthetic_fixtures=True testing switch, **off by default**; contract_reviewed is an operator assertion requiring independent documentary confirmation before configuration. No shipped provider has this assertion. A code flag is not independent legal permission, nor is a synthetic fixture rights grant permission for Google, Bing, Brave, or any other real upstream.
- First-party Web index admission is separate from the external-rights gate: source identity goreecloud_index, evidence marker goreecloud-local-sqlite-fts5, the exact

built-in OwnedIndexAdapter type, and owned_index rights must all match. No caller may register an external adapter with that reserved identity. The local adapter performs no outbound fetch and is limited to the Web category; a token that merely claims owned_index cannot authorize a network source.

- Up to eight named providers may be registered; each can return at most 40 SourceResult cards, with at most four concurrent calls **within one invocation**. The current Linux-native HTTPServer has one synchronous request worker and therefore limits fusion calls to one at a time in that *specific* Development process. It also bounds idle accepted client connections to five seconds. The dispatcher alone does not impose cross-process or multiworker concurrency budgets. A source receives the query only if admitted; inaccessible or denied sources are not called.
- The default cooperative per-provider deadline is three seconds (maximum ten), starting **after its semaphore slot is acquired** so a queued source is not timed out without a chance to execute. Real adapters **must** be asynchronous, honor cancellation, enforce their own socket/client deadline, upstream rate budget and egress policy, and never spawn unbounded blocking work. This module is not a process-level kill switch for a malicious or stuck third-party connector.
- Invalid adapter shape, mixed source identity, invalid ranks and unpublishable/unsafe result URLs fail that **one provider** with zero accepted cards instead of treating filtered cards as healthy results. This preserves other admitted providers' valid results. Source errors are counted, not logged or reflected with query-bearing exception messages; no hidden retries, substitute providers or user-query cache are implemented.
- FusionResponse has schema_version=1, category, a unified results tuple, source providers outcome tuples, and overall status (ok, partial, empty, unavailable). partial covers degraded or blocked configured providers; empty indicates permitted sources responded but produced no publishable results. Only the isolated native loopback adapter exposes this as Development JSON. This is **not** a public route or a promise of user-facing production behavior.
- The native fusion service may read the owner's local FTS5 index for Web; synthetic provider fixtures remain offline and opt-in. The currently deployed SearXNG-derived Development site, Caddy ingress, public host and source provider configuration are unchanged; a new unshipped .pyz candidate contains the native offline fusion route.

v0 deterministic behavior

- Every card records a validated lowercase source identifier, category, one-based position, HTTP(S) destination, title and optional description or image media URL; invalid destinations/cards are discarded. Category mismatch, malformed source identity, invalid rank or oversized input fails closed.
- Web and News use conservative destination URL equality, stripping fragments, default ports and listed tracking parameters. Videos also recognize a narrowly validated identical YouTube clip across watch, youtu.be, Shorts, live and embed URLs. Ambiguous IDs, nonstandard ports, mirrored hosts, syndication copies and separate news stories are **not** automatically merged. The original display destination is preserved.

- Images deduplicate by a validated original/media URL, never just the page hosting the asset. Images without media identity remain separate until richer evidence exists. Perceptual hashing is not implemented.
- A duplicate receives at most one vote per distinct source; if a single source returns that URL more than once, its best position wins. A score sums 1 / (60 + position) across its source votes (reciprocal rank fusion). Stable encounter order breaks exact-score ties.
- Source/rank evidence is internal to the model, not a provider-selection requirement in the product UI. Exposing or hiding attribution is subject to each approved provider agreement.
- Inputs are bounded to 1,000 cards, output to 100; no user query, response, logging, HTTP handler, SQLite table, cache or provider SDK is involved. Title (512 characters), description (2,000) and URL (4,096) fields are separately bounded before inclusion.

Integration acceptance still required

1. **License gate:** Only explicit, source-specific allowed usages can reach a provider adapter or populate the production fuse input. A *technically accessible* upstream HTML page is not contractual permission. Brave's derivative/caching restrictions and Google/Bing API lifecycle changes must be resolved before their native fusion activation.
2. **Adapter contract:** Map provider-specific payloads into typed vertical cards, validate redirects/media and language/locale, enforce per-provider limits, budget, timeout, network egress and content rights. Partial failures must not be disguised as an empty index.
3. **Orchestrator:** The offline cooperative dispatch contract exists, but it still needs a deployment-qualified async transport, signed and reviewed source-rights configuration, cross-request worker/request budget, upstream rate-limit/backoff policy and cancellation under real outages. Keep no invisible fallback source, expose degraded-state telemetry without raw user queries, and explain privacy implications of automatic provider fanout. Prefer POST for private Search to avoid query URLs.
4. **Ranking acceptance:** Compare fused results against a pinned synthetic and (separately authorized) real relevance corpus. Add query relevance, recency, quality/spam, domain diversity, language and visual/media similarity signals only after correctness and benchmark evidence.
5. **Runtime:** A real loopback-owned POST /v1/fusion exists for local-index Web results and optional synthetic Development data. It is not the deployed public Search service. Provide load, stalled-client, recovery, rate limiting, security, authentication, privacy, observability and rollback acceptance on the exact candidate before any public or supported-native-OS release claim.

Offline tests: python3 -m unittest discover -s tests/native -p test_fusion.py -v. Dispatch tests: python3 -m unittest discover -s tests/native -p test_dispatch.py -v. The candidate native Linux .pyz allowlist now includes the owned fusion, dispatch, sealed index-provider and synthetic-fixture modules. Local package tests and source-revision checks do not substitute for install, systemd lifecycle, signed-distribution, upgrade, real-provider, or live ingress acceptance.